MAPLE & MAIN · DECISION PRODUCTS

A KPI page that tells you when to care

Ten metrics, a seasonal baseline behind each one, and a ranked list of what actually needs a decision this week. Design notes for the dashboard.

Synthetic data. No client, employer or proprietary information.

EXECUTIVE SUMMARY

Most dashboards show every movement. This one shows the 10.8% that are real.

7 times

more flags would be raised by the usual "highlight anything over two per cent" rule than the data supports. That is how a page teaches people to ignore it.

9

items on this week's list, ranked, each written as a sentence with an owner attached. 5 of them need a decision; the rest are context.

5

models reporting on the same page, so it says what is about to happen as well as what has happened: churn, acquisition, personas, demand, media.

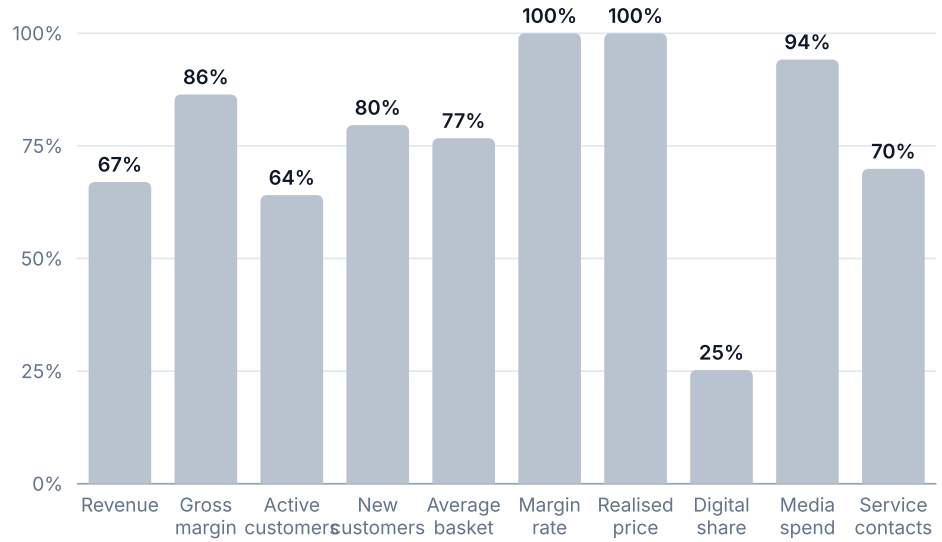
> **The design rule:** a number earns a place on the page only if someone owns it, a decision depends on it, and the page can say whether this week's movement is larger than its own normal variation.

THE PROBLEM

Why executives stop looking at dashboards

- **Everything moves every week.** Weekly retail numbers swing by a few per cent for no reason at all. A page that colours those swings red and green is reporting noise in two colours.
- **Seasonality is reported as news.** Garden furniture falls off a cliff every autumn. On a quarter-over-quarter view that looks like a crisis.
- **No one owns a number.** Forty tiles, no names, no decisions. The meeting becomes a tour of the page.
- **It only looks backwards.** The models that predict what happens next live somewhere else entirely.

How often each metric moved more than 2% in a week



Across 1,040 metric-weeks with at least a year of history behind them.

DESIGN

Four rules, applied to every tile on the page

1. A metric needs an owner and a decision

Each of the ten metrics is on the page because a named person uses it to decide something. "Realised price" exists so merchandising can see whether the discount plan is holding, not because it is available.

2. Signal before size

Every number is compared with what it should have been for that week of the year, not with last week. The page says "within normal" or flags it, before it says how big the change was.

3. Sentences, not charts, at the top

The first thing on the page is a ranked list of what needs attention, written in plain English with the owner attached. Charts are for the person who wants to check.

4. The models report here too

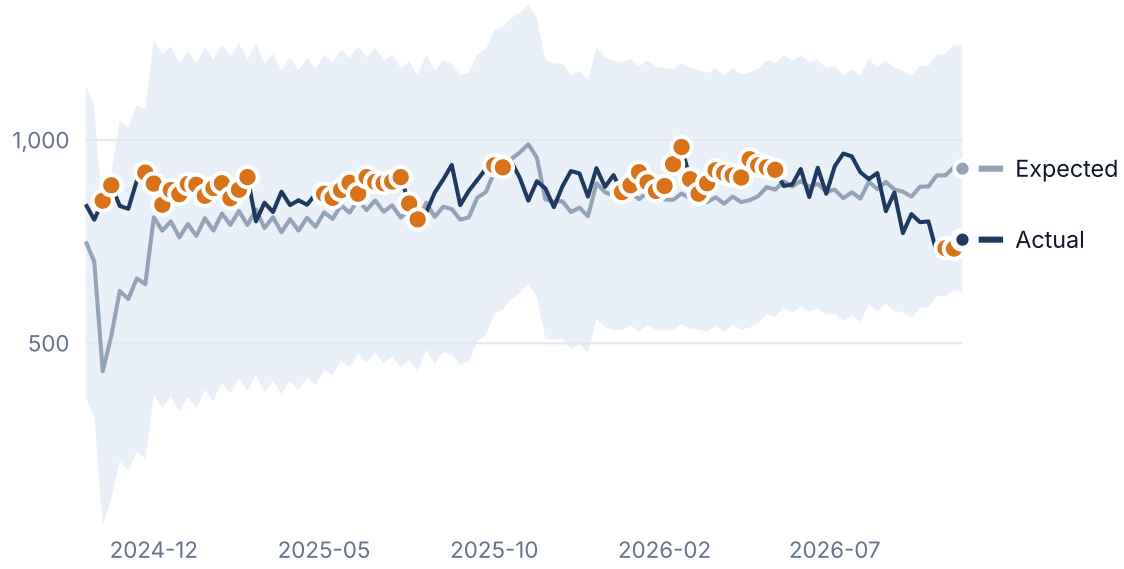
Churn risk, acquisition value, persona coverage, forecast accuracy and media return appear on the same page, each with the action it implies.

METRIC	OWNER	THE DECISION IT SUPPORTS
Revenue	Chief executive	Is the business on plan this quarter?
Gross margin	Finance	Is growth being bought with discount?
Active customers	Marketing	Is the base growing or just spending more?
New customers	Acquisition	Is acquisition keeping pace with lapsing?
Average basket	Merchandising	Are people buying more per trip?

THE METHOD

Every metric gets a baseline, and the baseline never sees the week it judges

Service contacts: weekly, against its seasonal baseline



Shaded band: three standard deviations around the expected level. Amber points: weeks the control rules flagged.

For each metric and each week, a model is fitted on **everything before that week only**: a trend, an annual shape, the holiday peak, back-to-school and flyer weeks. It produces an expected level and a measure of how much a normal week varies around it.

Three rules then decide whether to speak up, borrowed from a century of factory process control:

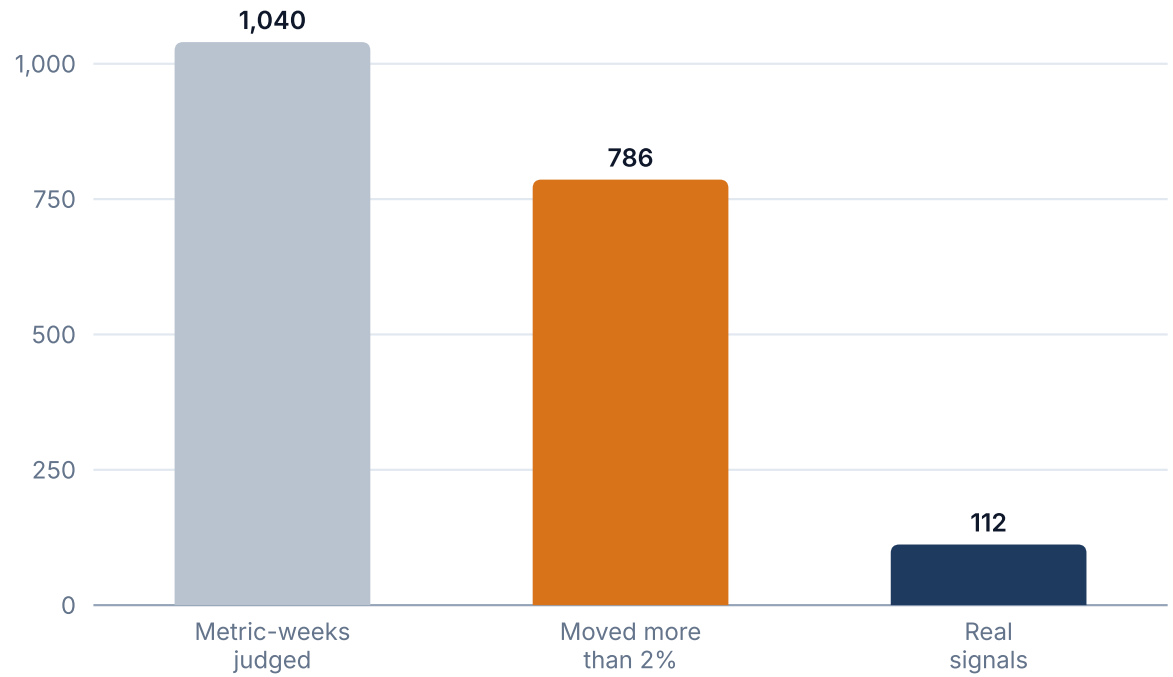
- a week beyond **three** standard deviations
- **two of three** weeks beyond two
- **eight** weeks in a row on the same side of the baseline

> The third rule is the one that earns its keep. A drift of half a standard deviation never trips an eyeball, and it is exactly how a problem arrives.

THE AUDIT

Of 786 movements a normal dashboard would highlight, 112 are real

Metric-weeks, by how they would be treated



Ten metrics across 104 weeks each. A signal is a week that broke one of the three control rules.

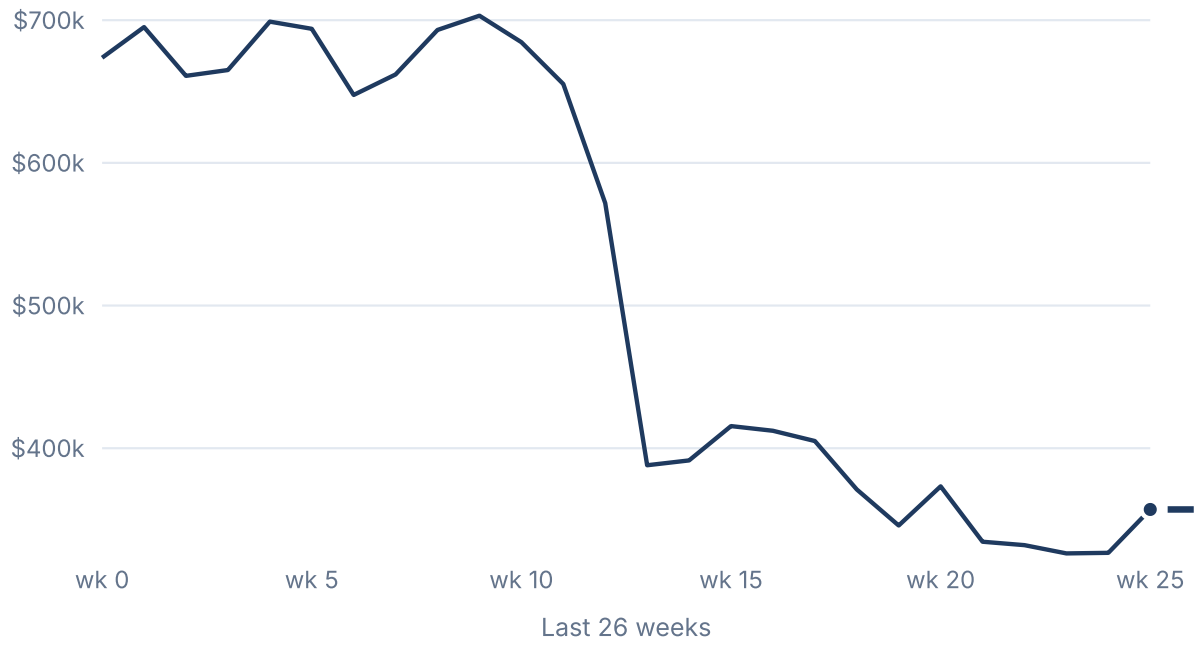
METRIC	WEEKS WITH A SIGNAL	SHARE OF WEEKS
Service contacts	51	49%
Realised price	19	18%
Margin rate	10	10%
Revenue	8	8%
Gross margin	7	7%
Active customers	4	4%

Service contacts and realised price are genuinely drifting and deserve the attention. Revenue, margin and digital share mostly do what they always do. A page that shouted equally about all ten would bury the two that matter.

THE EXAMPLE THAT MAKES THE CASE

Outdoor and garden is down 45% on the quarter, and nothing is wrong

Outdoor & Garden weekly revenue



It is October. The category does this every year.

A standard report compares the last quarter with the one before and shows a **45%** fall. In most companies that produces a meeting.

Measured against its own seasonal pattern, the category is **0.1 standard deviations** from where it should be, which is to say exactly where it should be.

Outdoor & Garden is down 45% and that is normal

Against its own seasonal pattern it is 0.1 standard deviations from where it should be. No action needed.

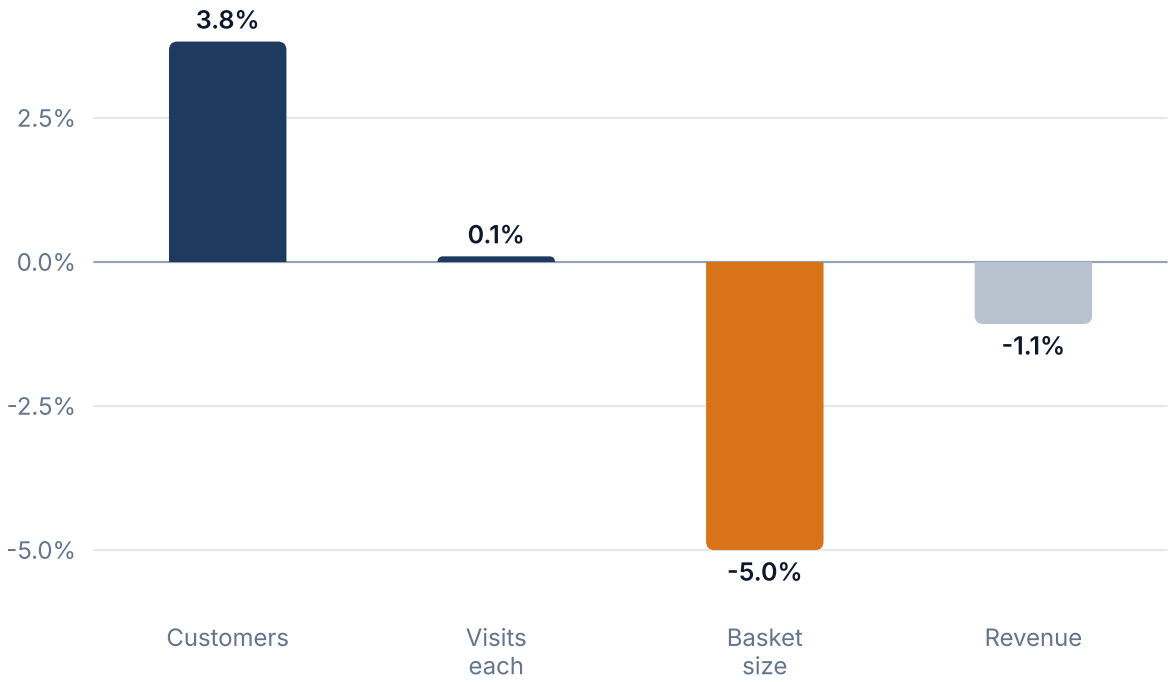
NOT A PROBLEM · CATEGORY · COMMERCIAL

The dashboard says so out loud, because the most valuable thing a page can do is take an item off the agenda.

THE FIRST FOLLOW-UP QUESTION

When revenue moves, the page already knows what moved it

Contribution to the change in revenue



Revenue is customers multiplied by how often they buy and how much they spend each time. The three parts add to the total.

PERIOD	REVENUE	CUSTOMERS	VISITS EACH	BASKET
Last 4 weeks	-9.3%	-6.1%	-0.5%	-2.6%
Last 13 weeks	-1.1%	3.8%	0.1%	-5.0%
Last 52 weeks	**15.6%**	**14.6%**	**0.7%**	**0.3%**

Over the year the business grew +15.6%, almost entirely by having more customers. Over the last quarter revenue is flat because a +3.8% rise in customers was cancelled by a 5.0% fall in basket. Those are two completely different conversations, and the page separates them before anyone asks.

THE ALERT ENGINE

Nine sentences, ranked, each with a name on it

Average basket is 5.3% down on the quarter

The last 13 weeks averaged $176 against $186 in the 13 before. The gap is larger than normal week-to-week variation would pro...

SUSTAINED CHANGE · MERCHANDISING

New customers is 10.0% up on the quarter

The last 13 weeks averaged 708 against 644 in the 13 before. The gap is larger than normal week-to-week variation would produ...

SUSTAINED CHANGE · ACQUISITION

SK is behaving unusually for the time of year

SK revenue is 2.9 standard deviations above its own seasonal baseline (two of three weeks beyond two standard deviations). It...

WHERE IT IS HAPPENING · COMMERCIAL

Service contacts is below its seasonal baseline

Service contacts is 8 weeks in a row on one side of the baseline. The latest week is 755 against an expected 930. This is a p...

CONTROL SIGNAL · CUSTOMER SERVICE

Items come from four places: a control rule firing, a quarter-on-quarter change larger than normal variation, a slice behaving unusually for the season, and the models.

Each is scored on how far it is from normal and how much money sits behind it, then sorted. Severity is a colour, not a word count.

- **High**: beyond three standard deviations
- **Medium**: a pattern, or a real change in the wrong direction
- **Low**: a real change in the right direction
- **Green**: something that looks alarming and is not

> The list is capped by the data, not by a target length. A quiet week produces a short list, which is the correct output for a quiet week.

FORWARD-LOOKING

Five models, reporting on the page the executive already reads

Customers at risk
Project 1: churn and retention value

$5,159,840

12-month margin at risk

Rank the monthly offer list by value at risk, not by risk alone.

Acquisition
Project 2: acquisition propensity

$554,255

First-year margin available a year

Order the outreach list by the propensity scorecard before the next campaign.

Who we can personalise to
Project 3: segmentation and personas

19%

of active customers carry a persona

Extend the purchase window to two years before the next content refresh.

Forecast against actual
Project 4: demand forecasting

1.9%

error over the last 13 forecast weeks

Order to the 63% service level, not to the forecast itself.

Media
Project 5: marketing mix model

$523,123

a year from reallocating the same budget

Run two-depth holdouts before acting on any channel-level number.

Each panel is the output of a separate project in this portfolio. Putting them on the KPI page is the difference between a reporting tool and a decision product: the page says what happened, what is about to happen, and what to do about it, in one place.

HONEST LIMITS

What this page cannot do

- **It cannot tell you why.** A control signal says a metric left its normal range. The cause is a separate question, and usually needs an experiment rather than another chart.
- **It cannot prove a campaign worked.** Sales rose and marketing spent money. Only a holdout settles which caused which, which is why projects 1 and 5 are built around them.
- **The baseline is only as good as the history.** A metric needs a year before it can be judged, and a genuine step change will be called a signal for several weeks before the baseline catches up. That is the correct behaviour, but it has to be explained once.
- **Three rules on ten metrics will produce false alarms.** Roughly one week in a hundred will trip a rule by chance. With ten metrics that is about one spurious flag a quarter, and the page is better for accepting it.
- **Nothing here replaces a human.** The alert list is an agenda, not a decision. Its purpose is to make sure the meeting starts on the two things that matter rather than the first chart on the page.

> Governance: every metric has one definition, written in SQL in one place. If the dashboard and the finance pack disagree, one of them is not using this mart.

APPENDIX

Method, definitions and limits

Data. One weekly fact table built from every order line, plus the media plan, the trading calendar and service records. 156 weeks to 2026-10-26. Slices by category, channel and province reconcile to the company total to the dollar.

Baseline. For each metric and week, a linear model on trend, three annual harmonics, holiday peak, back-to-school and flyer weeks, fitted only on earlier weeks. Metrics need 52 weeks of history before they are judged.

Control rules. Three standard deviations; two of three beyond two; eight consecutive weeks on one side.

Period comparison. The average of the last N weeks against the N before, with a standard error from the week-to-week variation of both. "Real" means the difference exceeds two standard errors.

Decomposition. Revenue is split into customers, orders per customer and average basket on a log scale, so the three parts add to the total change.

New customers are counted in the week they joined the loyalty programme, not the week of their first order: the latter puts every pre-existing member into the first week of the extract and invents an acquisition spike.

Limits. No competitor, weather or macroeconomic inputs exist in this dataset. The baseline treats a step change as a run of signals until it has a year of new history. False alarm rate is roughly one metric-week in a hundred.

Synthetic data. Maple & Main is fictional. Figures illustrate the method, not any real company.

Code. SQL, Python and SAS, with a metric dictionary and QC log, are in the project repository.